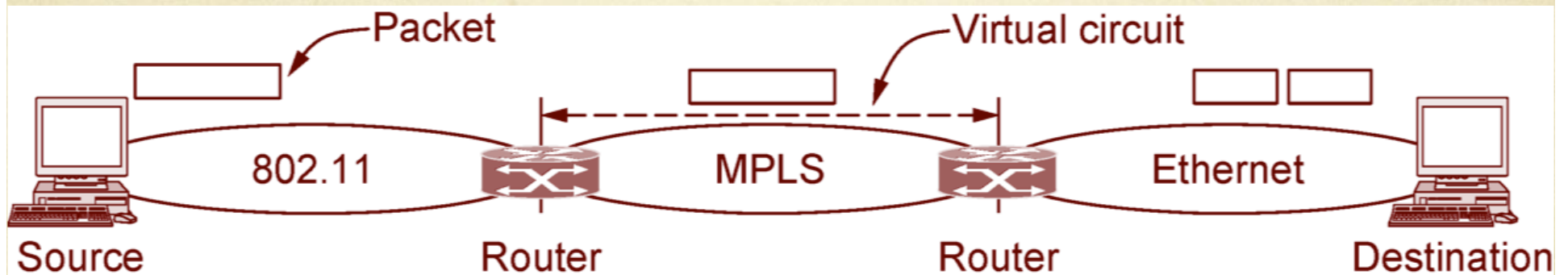


Internetworking

- ◆ Several routing algorithms considered so far assume
 - ◆ *Homogeneous* network
 - ◆ All nodes use common protocols in all layers
- ◆ *Not* true in reality
 - ◆ Networks may differ in several ways...

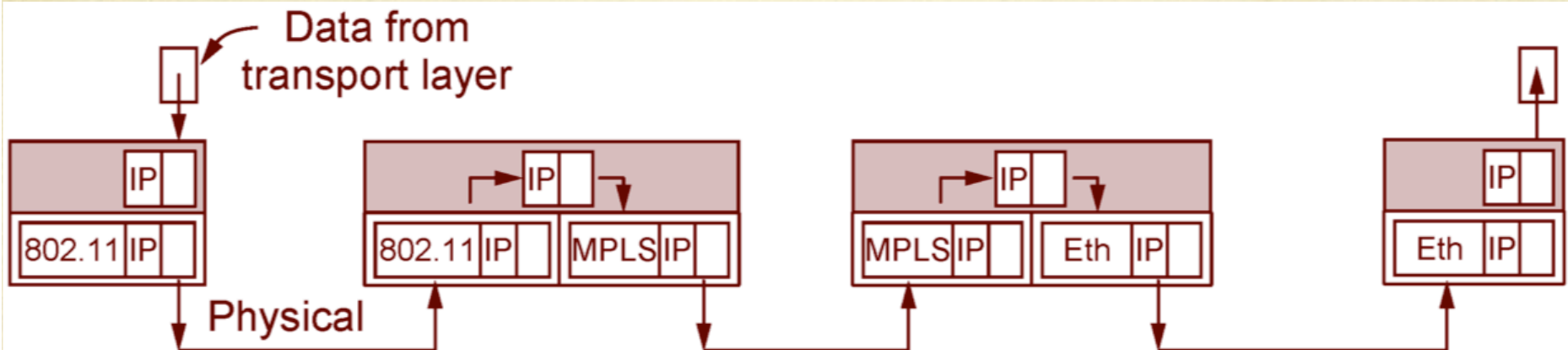
Internetworking

- ◆ In what ways do networks differ?
 - ◆ *Services*: connectionless versus connection oriented
 - ◆ *Addressing*: different sizes, flat or hierarchical
 - ◆ *Packet size*: different maximum sizes
 - ◆ *Quality of Service*: priority,



One solution – Cerf & Kahn

- ◆ *Common* network layer above *different* physical networks
 - ◆ E.g., Internet Protocol (*IP*): foundation of the *Internet*

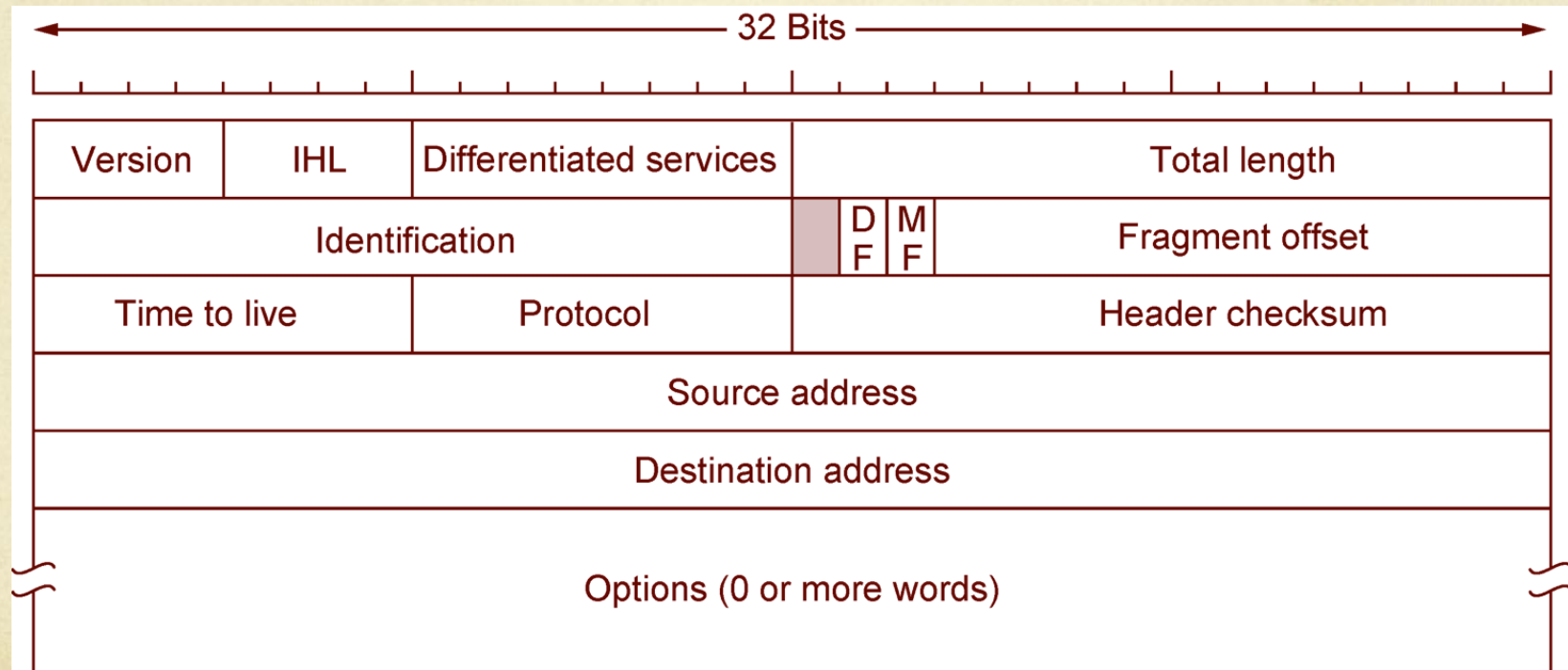


Internet Protocol (*IP*)

- ◆ Common network layer protocol
 - ◆ Glue that holds *Internet* together
- ◆ *Connectionless, best effort* service
 - ◆ No reliability guarantees
 - ◆ No ordering guarantees
- ◆ IP versions
 - ◆ Version 4 (*IPv4*)
 - ◆ Version 6 (*IPv6*)

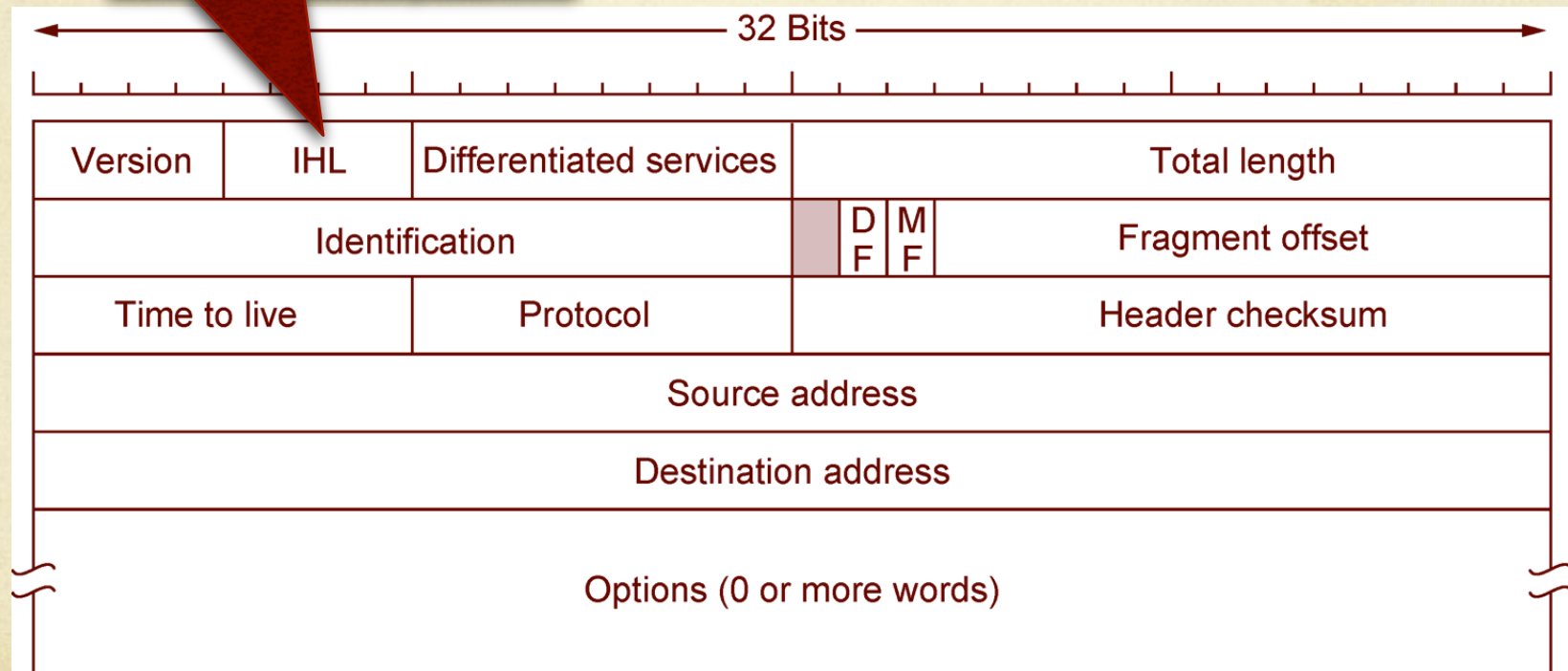
IP version 4 (*IPv4*)

- ◆ *IPv4 header* included in all network layer packets



IP version 4 (*IPv4*)

- Header length in 32-bit words – min 5 words (20 bytes), max 15 words (60 bytes)
- Used in all network layer packets

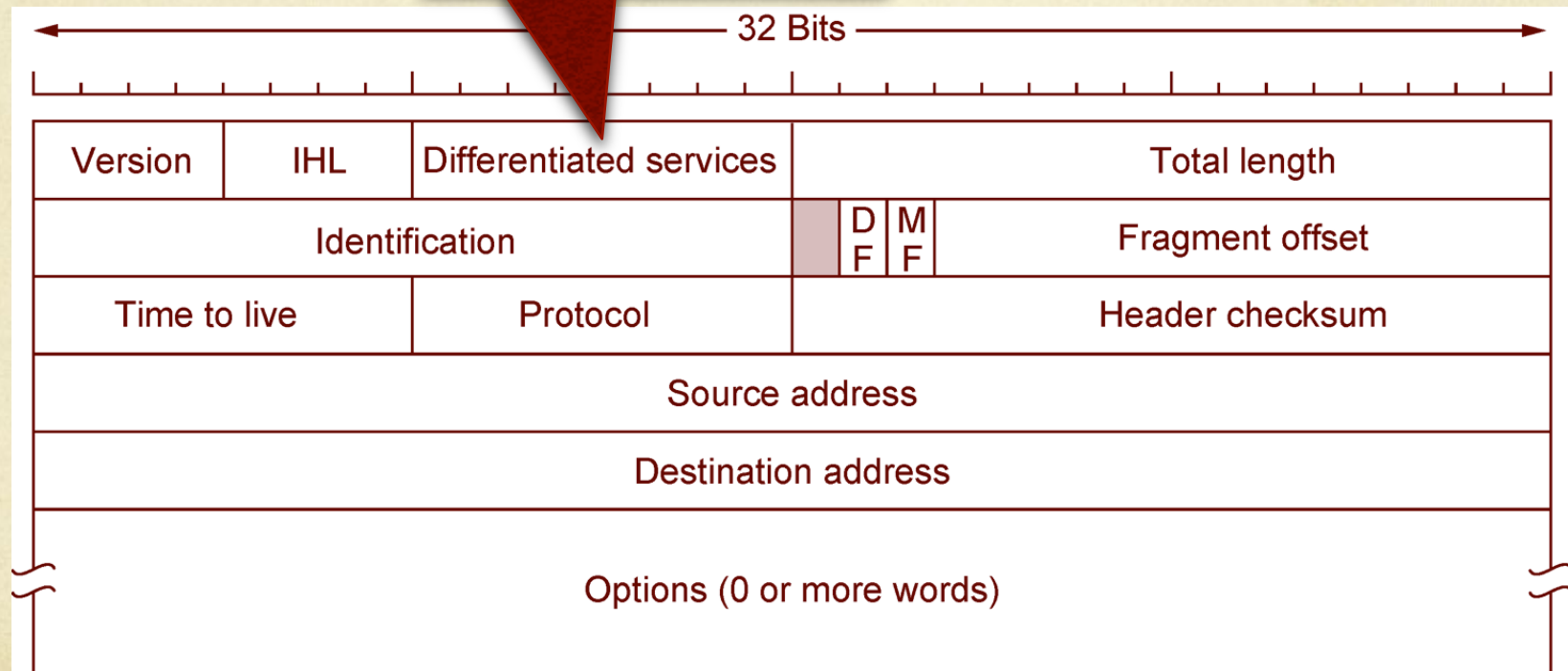


IP version 4 (*IPv4*)

◆ IPv4 *header*

*Current usage:
Service class & explicit
congestion notification
bits*

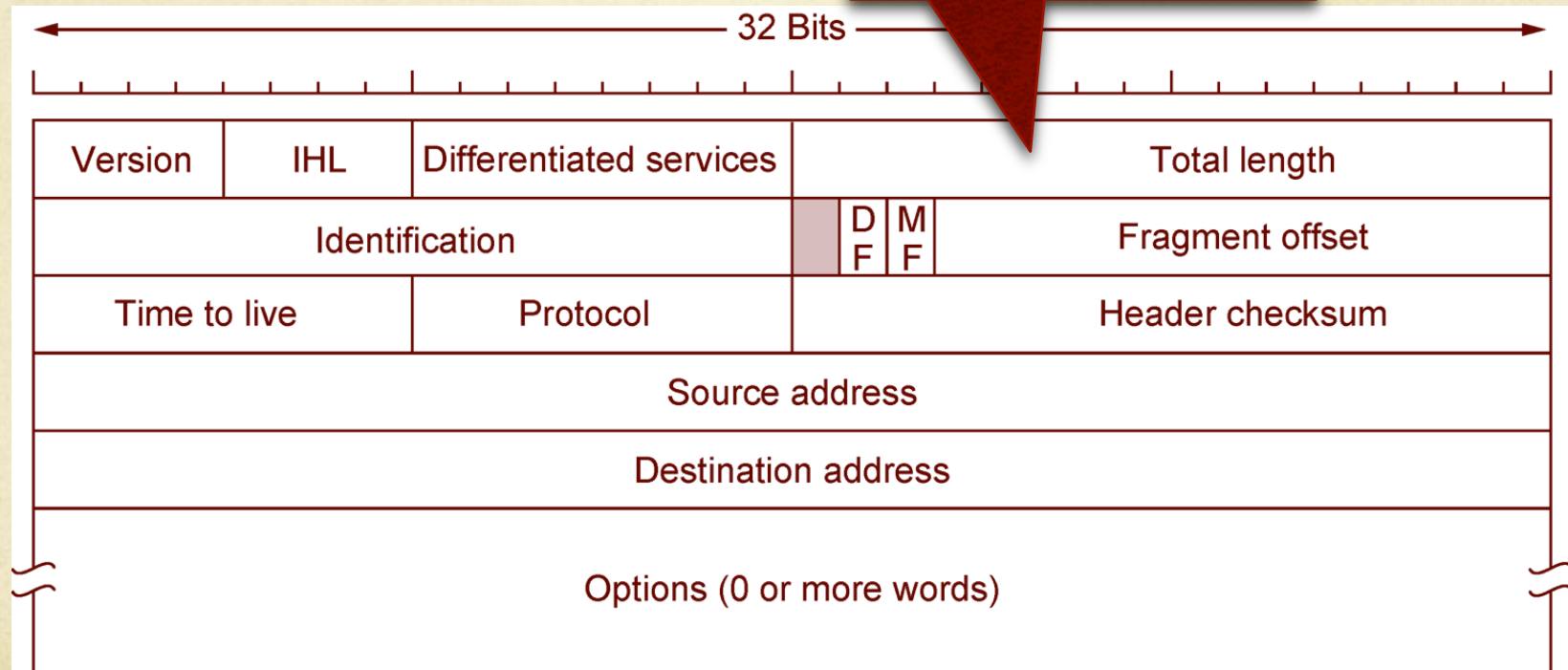
network layer packets



IP version 4 (*IPv4*)

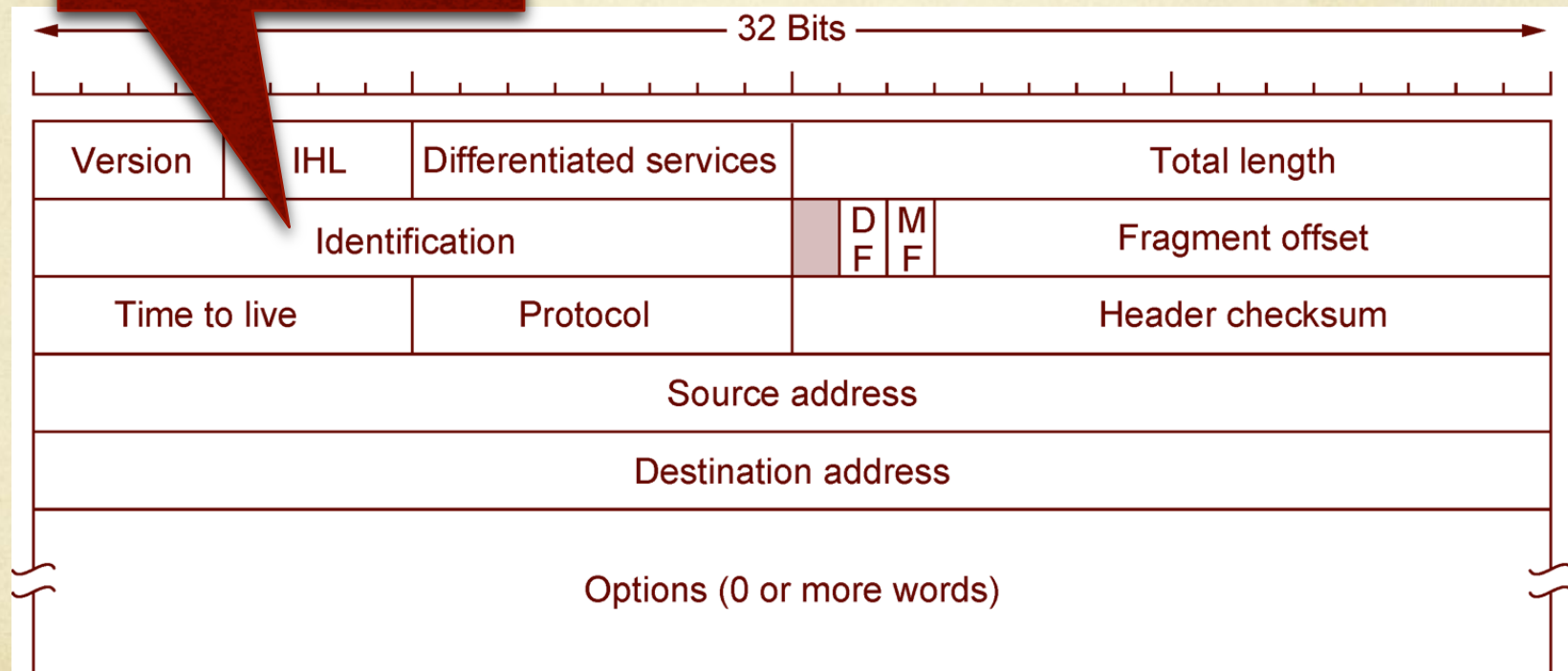
- ◆ IPv4 *header* included in all packets

Total length of entire packet (header + data) – max 64KB



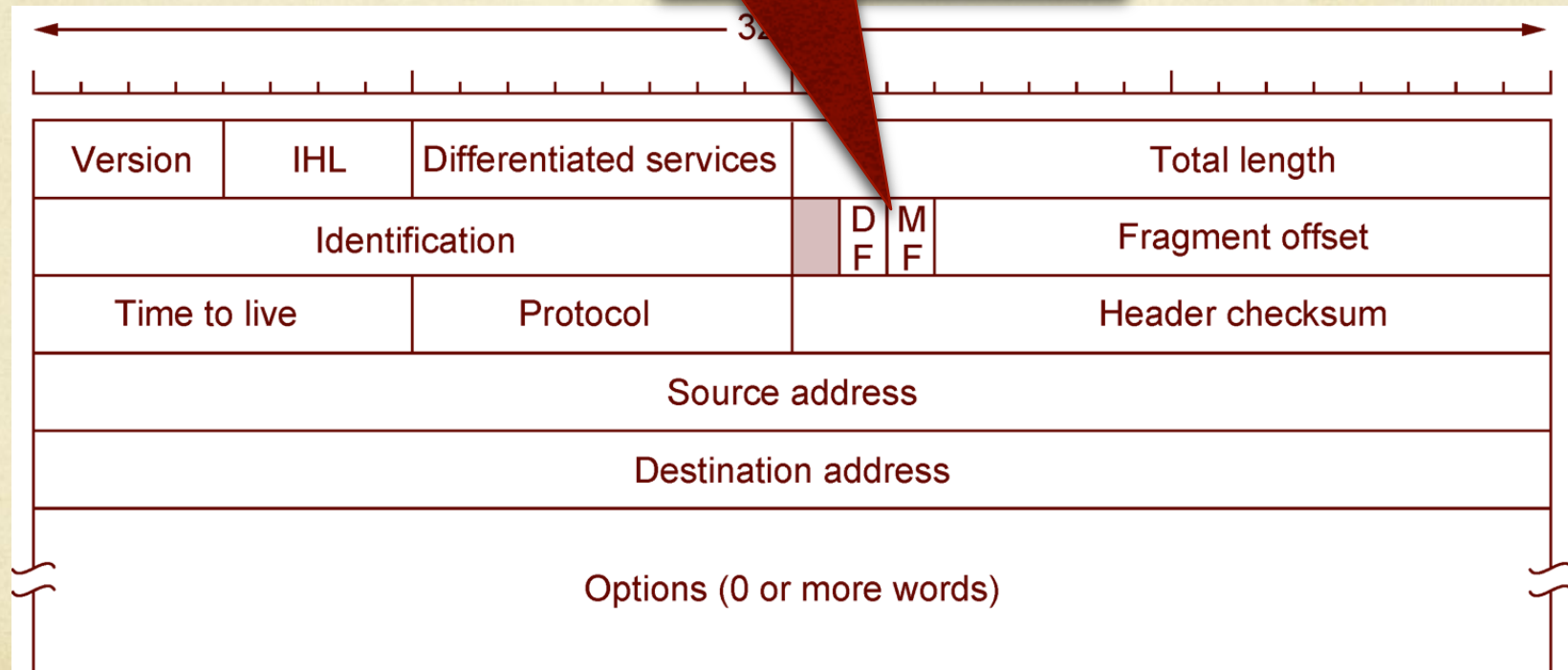
IP version 4 (*IPv4*)

- ◆ *Packet identification when fragmentation is used* included in all network layer packets



IP version 4 (*IPv4*)

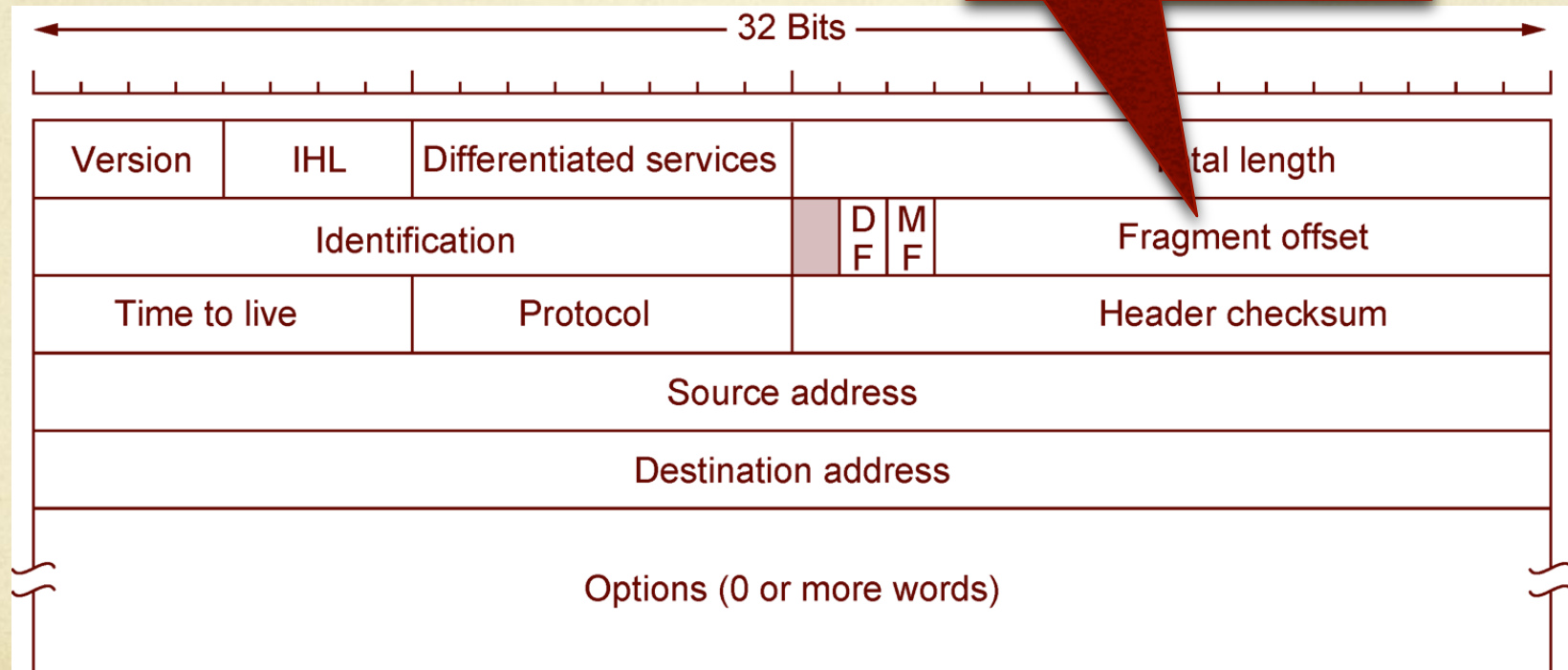
- ◆ IPv4 *header* included “Don’t Fragment” & “More Fragments” flags



IP version 4 (*IPv4*)

- ◆ IPv4 *header* included in all network packets

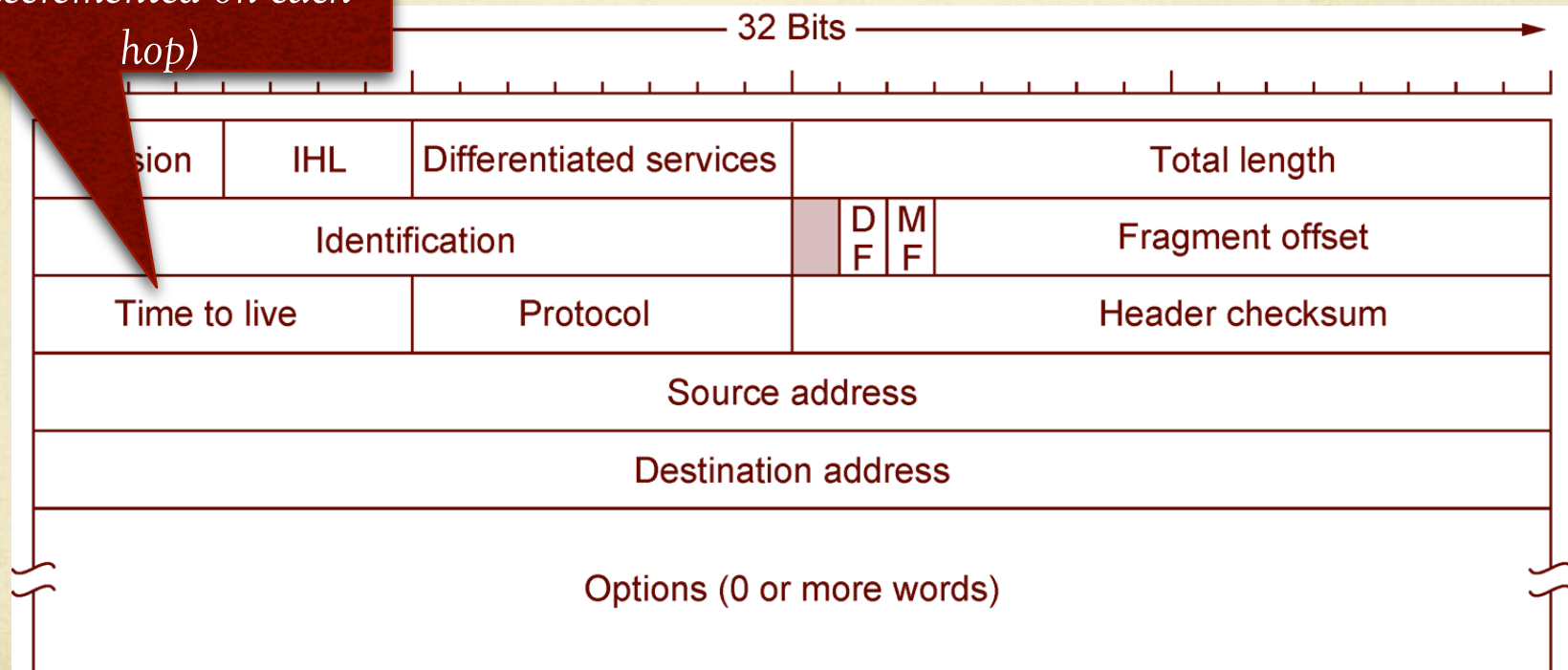
Fragment offset within packet (max 8K fragments per packet)



IP version 4 (*IPv4*)

Counter used to limit
packet lifetime
(decremented on each
hop)

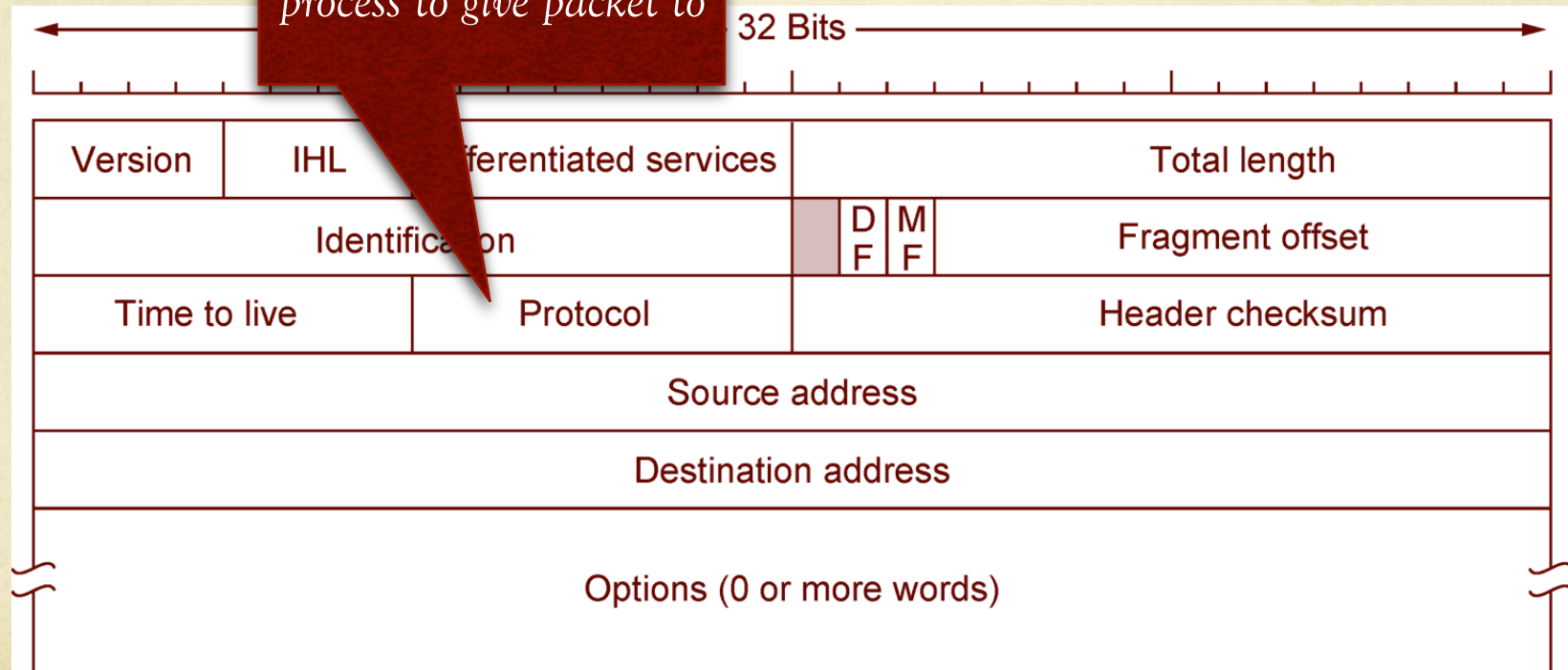
included in all network layer packets



IP version 4 (*IPv4*)

- ◆ IPv4 header is present in all network layer packets

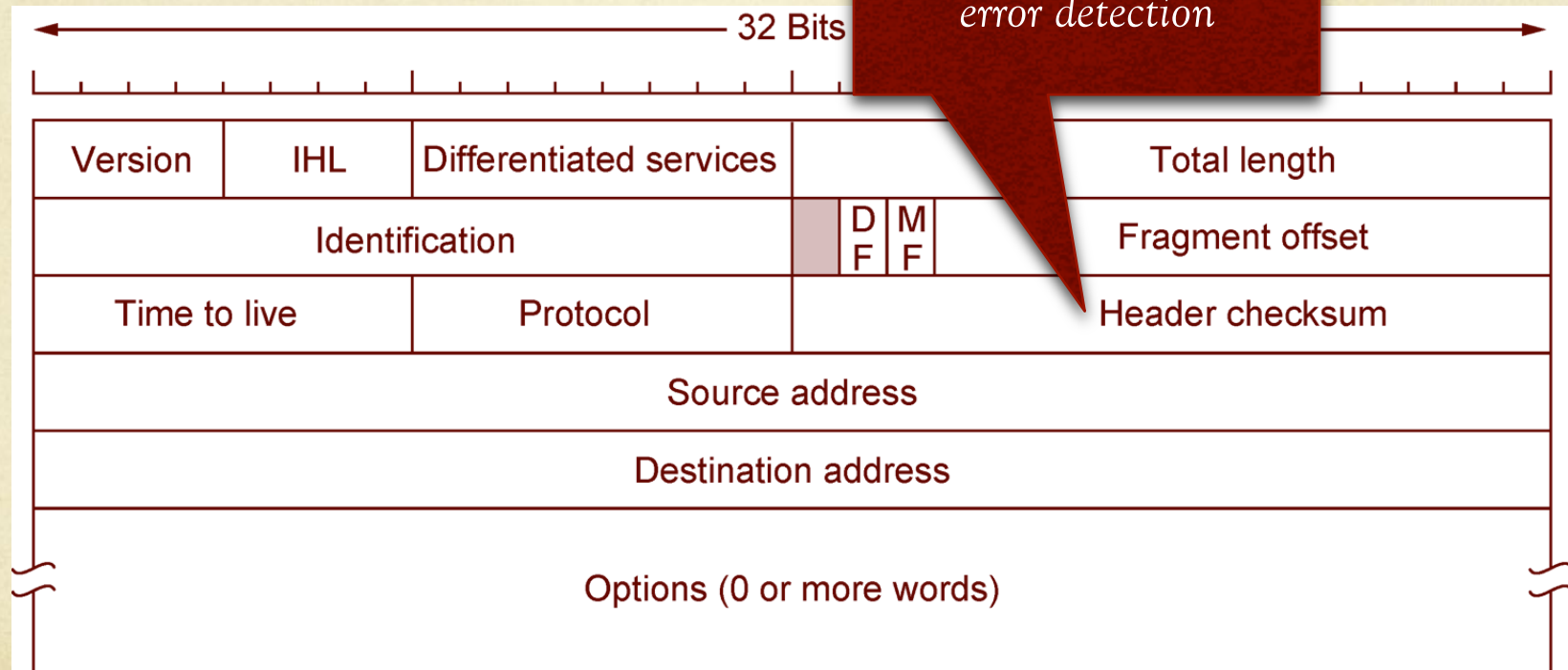
Which transport layer process to give packet to



IP version 4 (*IPv4*)

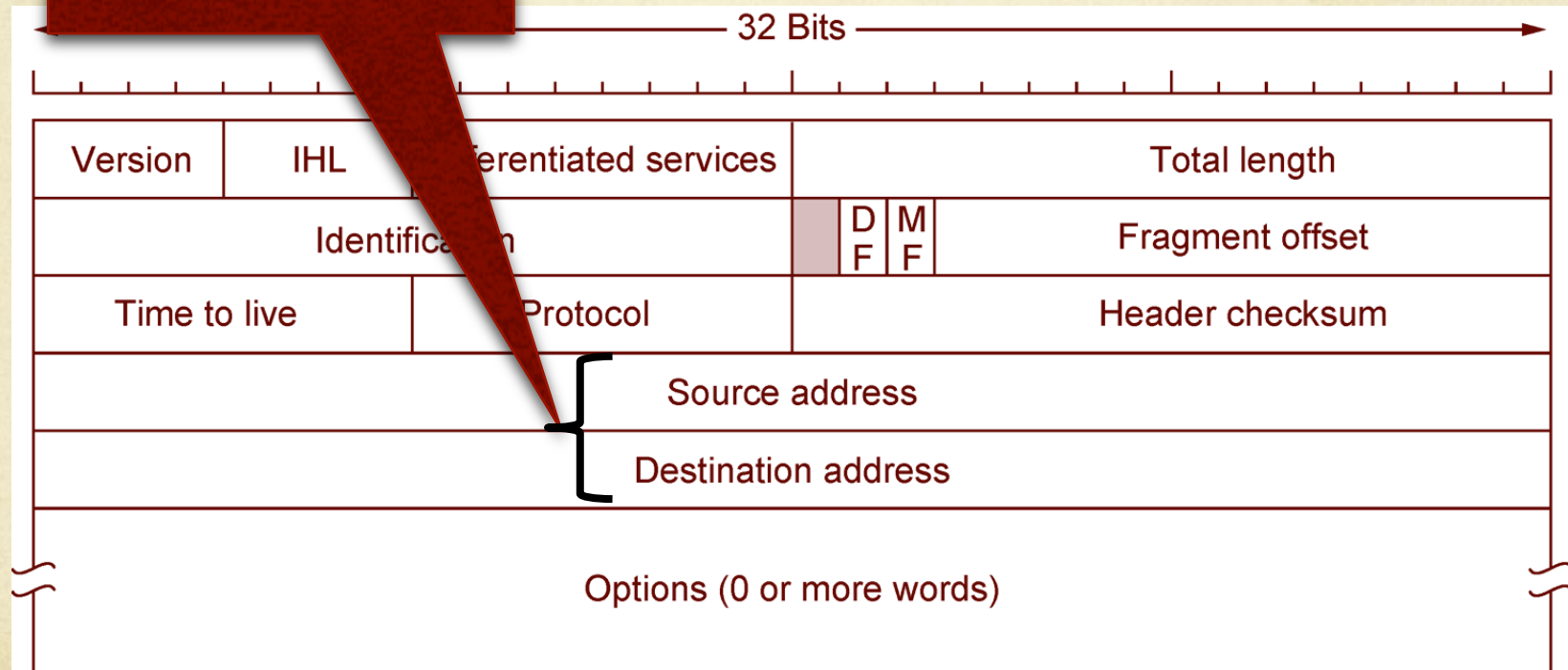
- ◆ *IPv4 header* included in all packets

Checksum used for
error detection



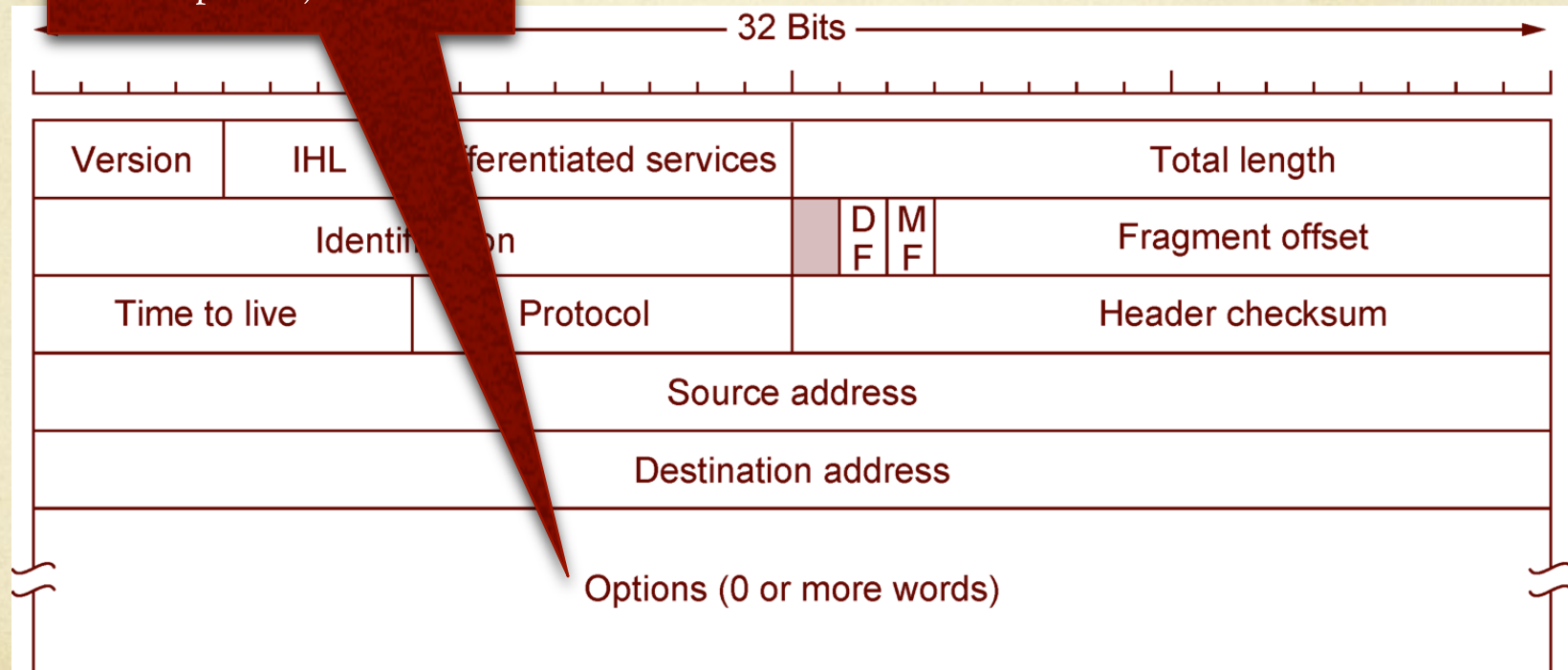
IP version 4 (*IPv4*)

- ◆ *Source & destination endpoint addresses* included in all network layer packets



IP version 4 (*IPv4*)

- Specify options – included in all network layer packets
security options, routing options, etc.

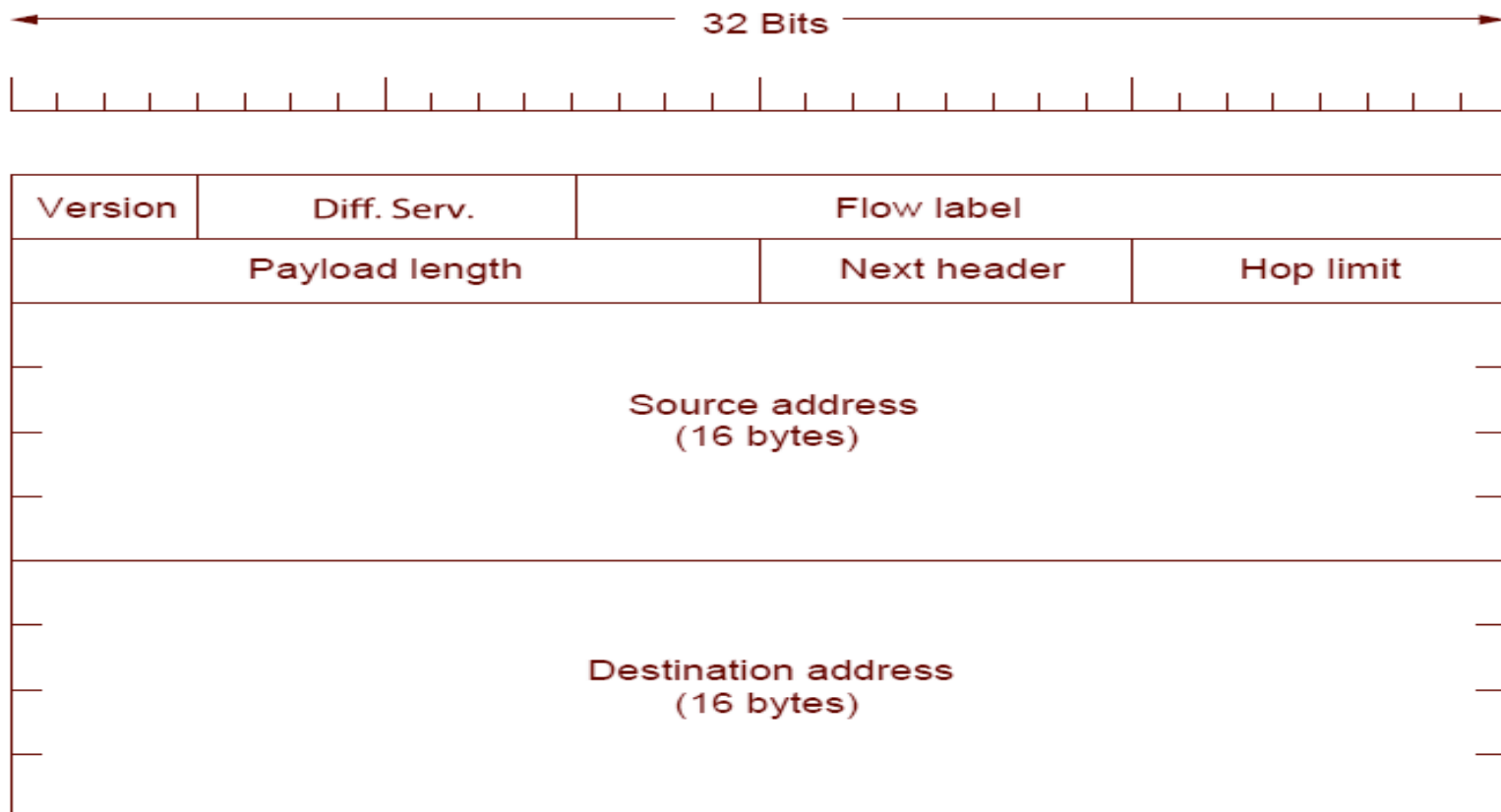


Keeping up with growth

- ◆ IPv4 can't keep up with *growing size* of the Internet
- ◆ New version → IPv6
 - ◆ Supports *128-bit* address
 - ◆ Has *simpler* header → reduces processing overhead
 - ◆ Better support for *options*
- ◆ Has not been easy to deploy on Internet-scale
 - ◆ Does not internetwork easily with IPv4!

IP version 6 (*IPv6*)

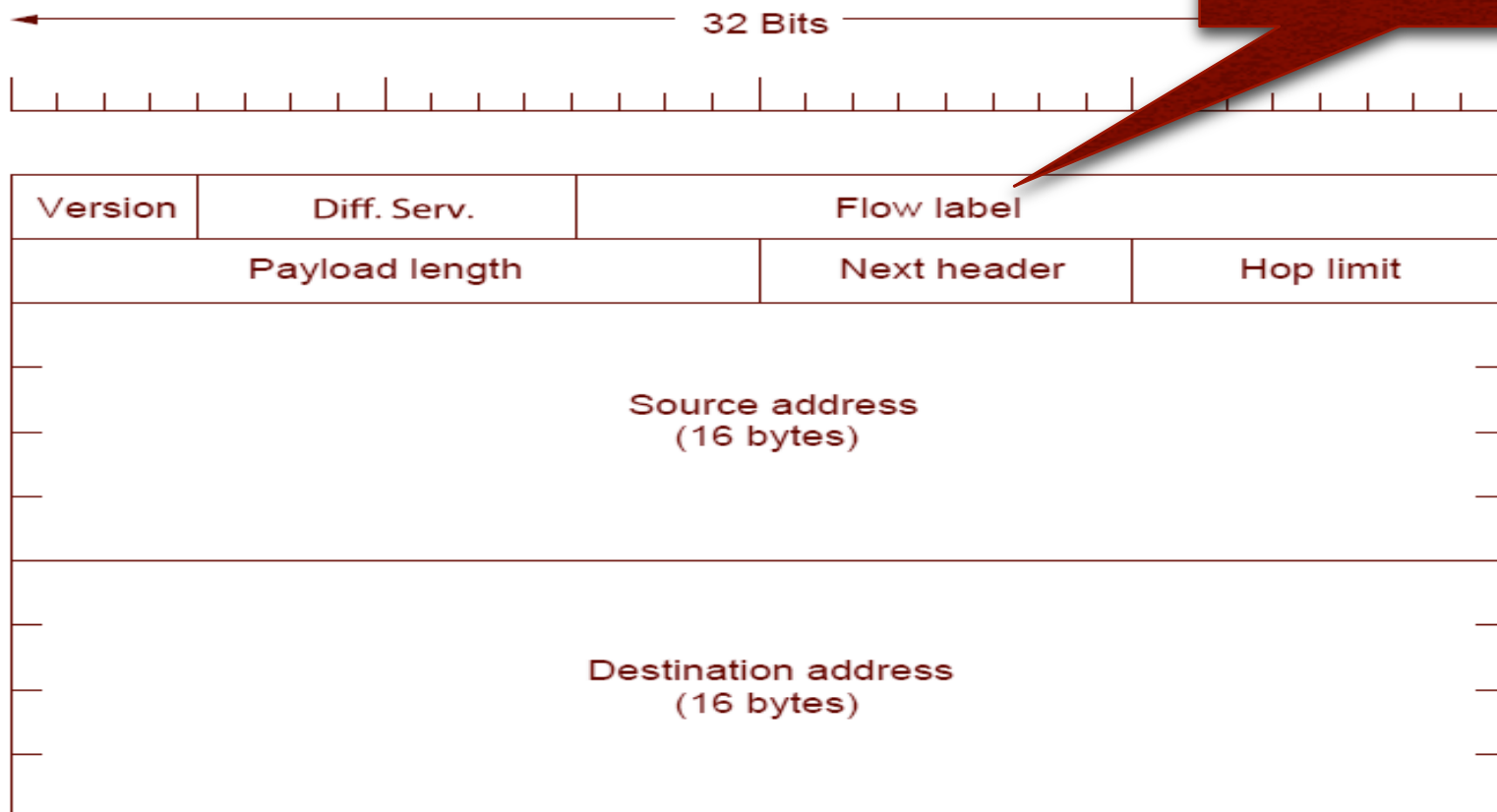
◆ *Required* header



IP version 6 (*IPv6*)

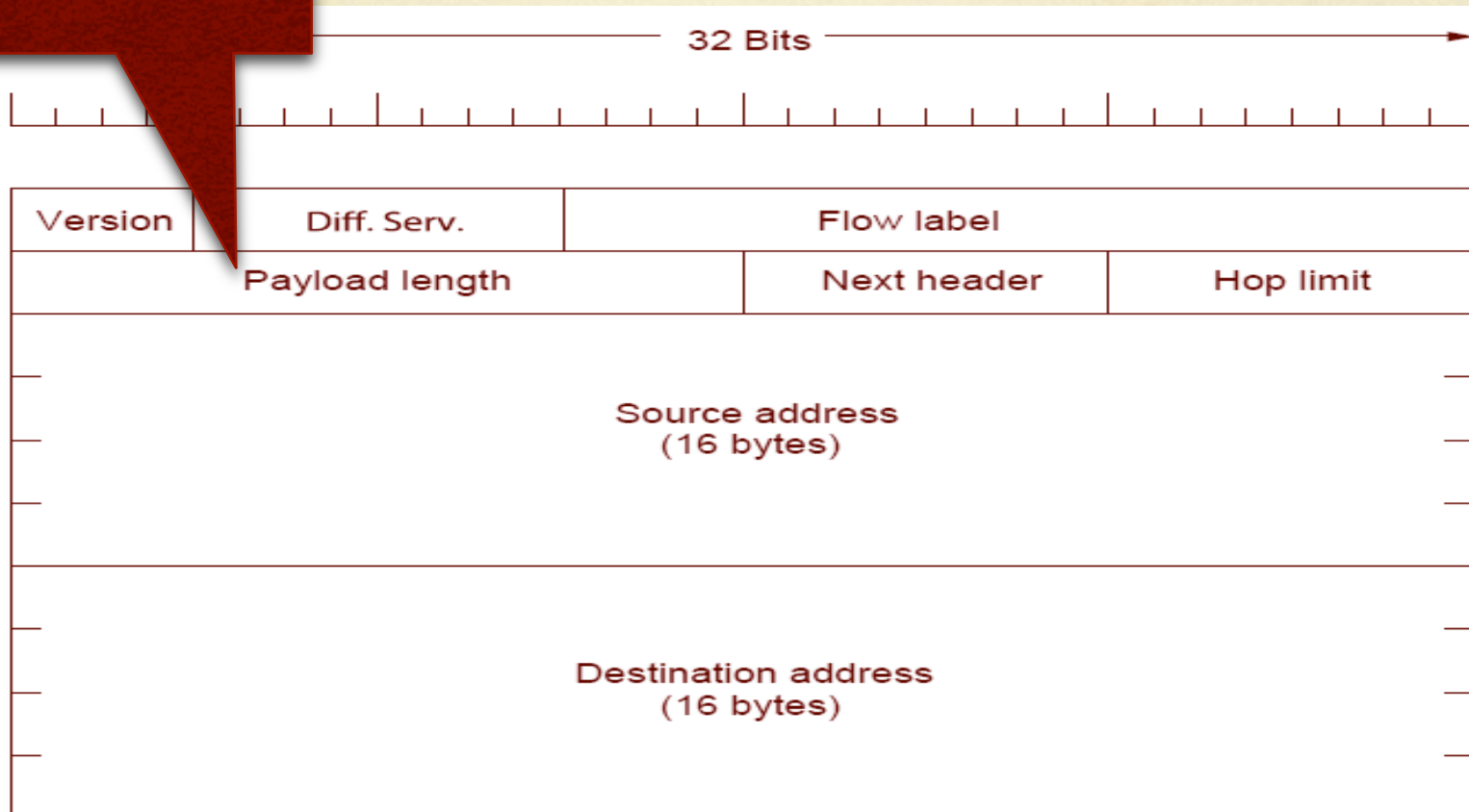
- ◆ *Required* header

Way to make groups of packets with same requirements



IP version 6 (*IPv6*)

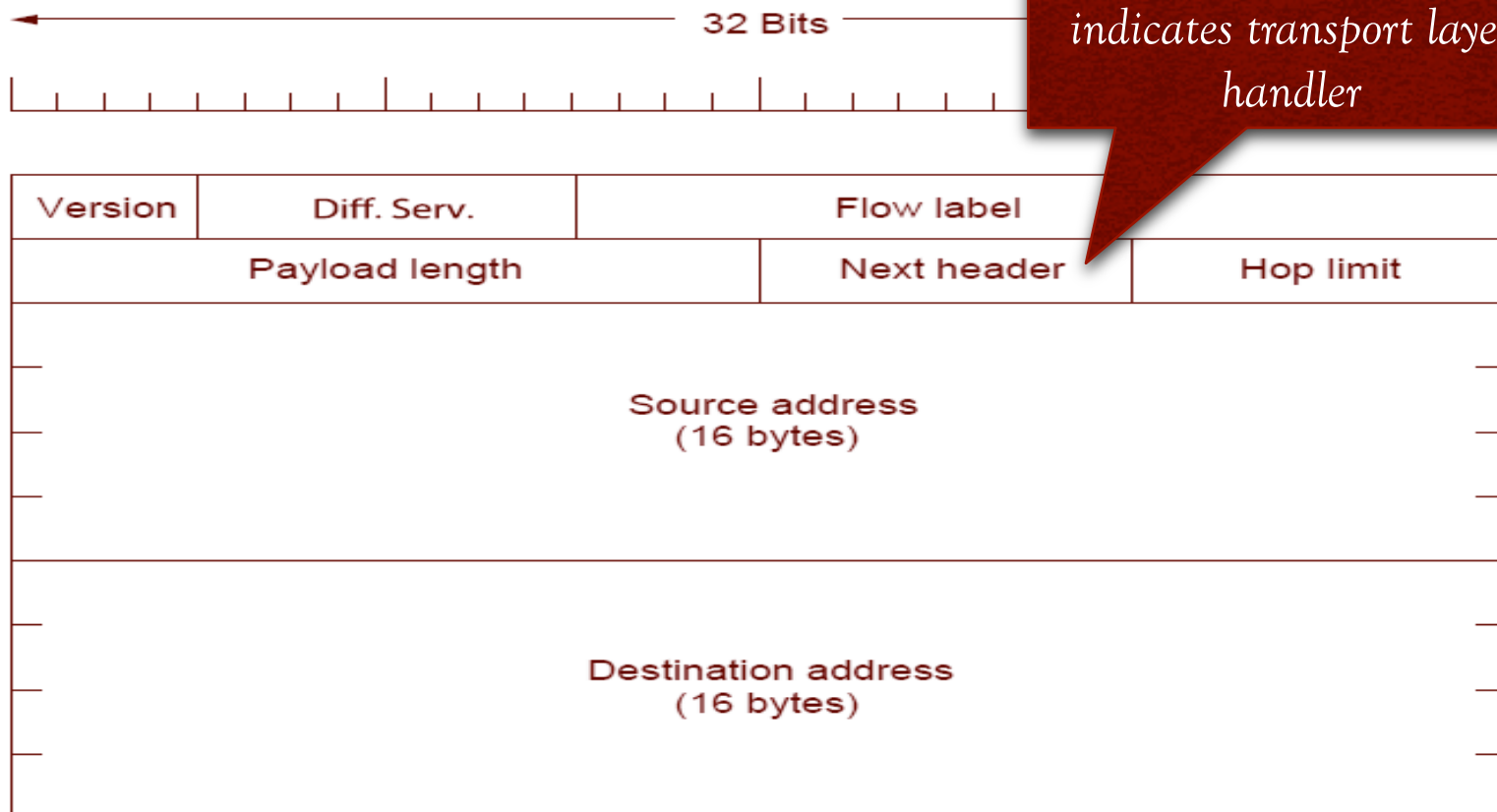
*Length of payload
after required header*



IP version 6 (*IPv6*)

◆ *Required* header

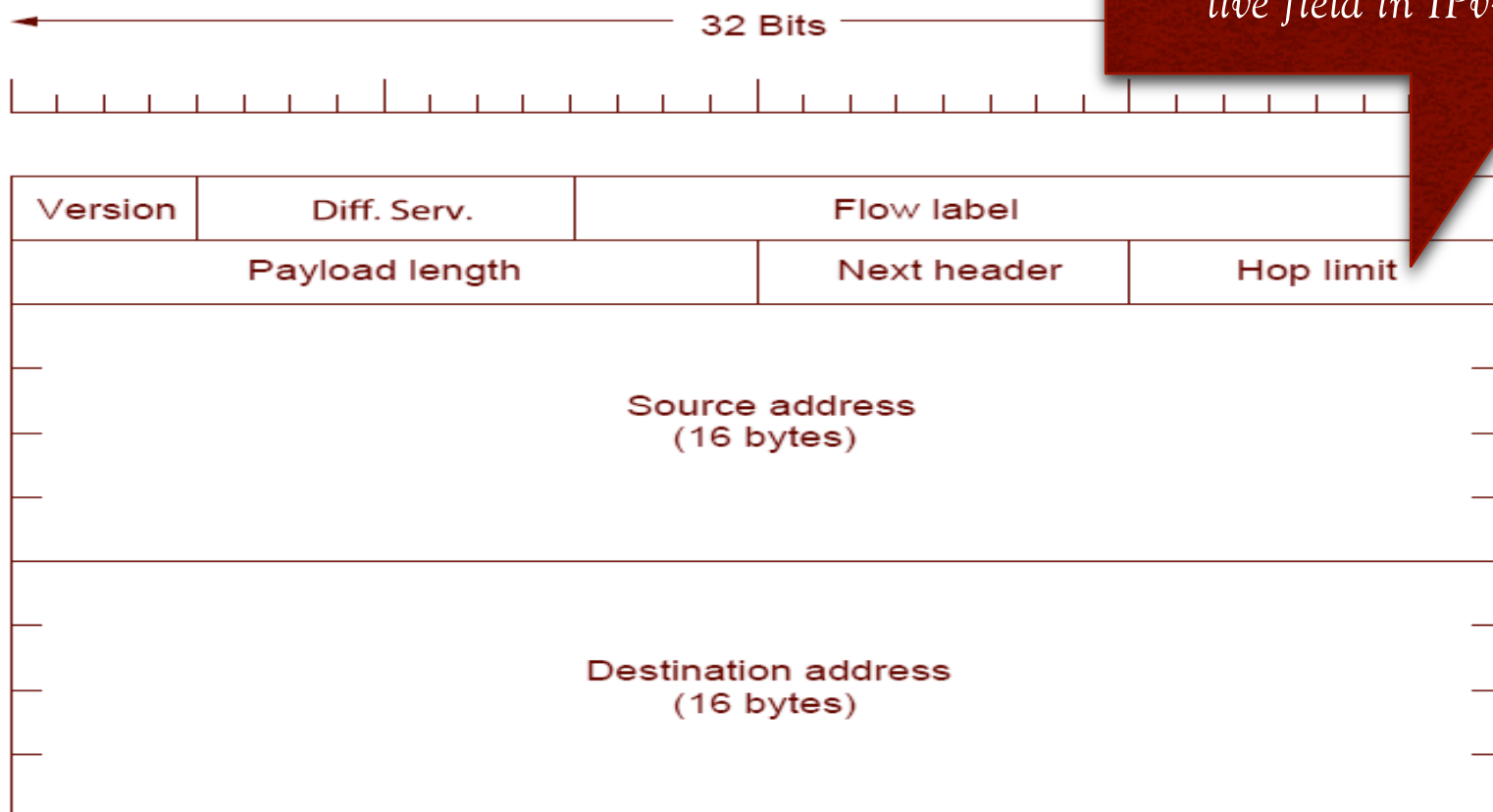
Indicates optional extension headers; in last extension header, indicates transport layer handler



IP version 6 (*IPv6*)

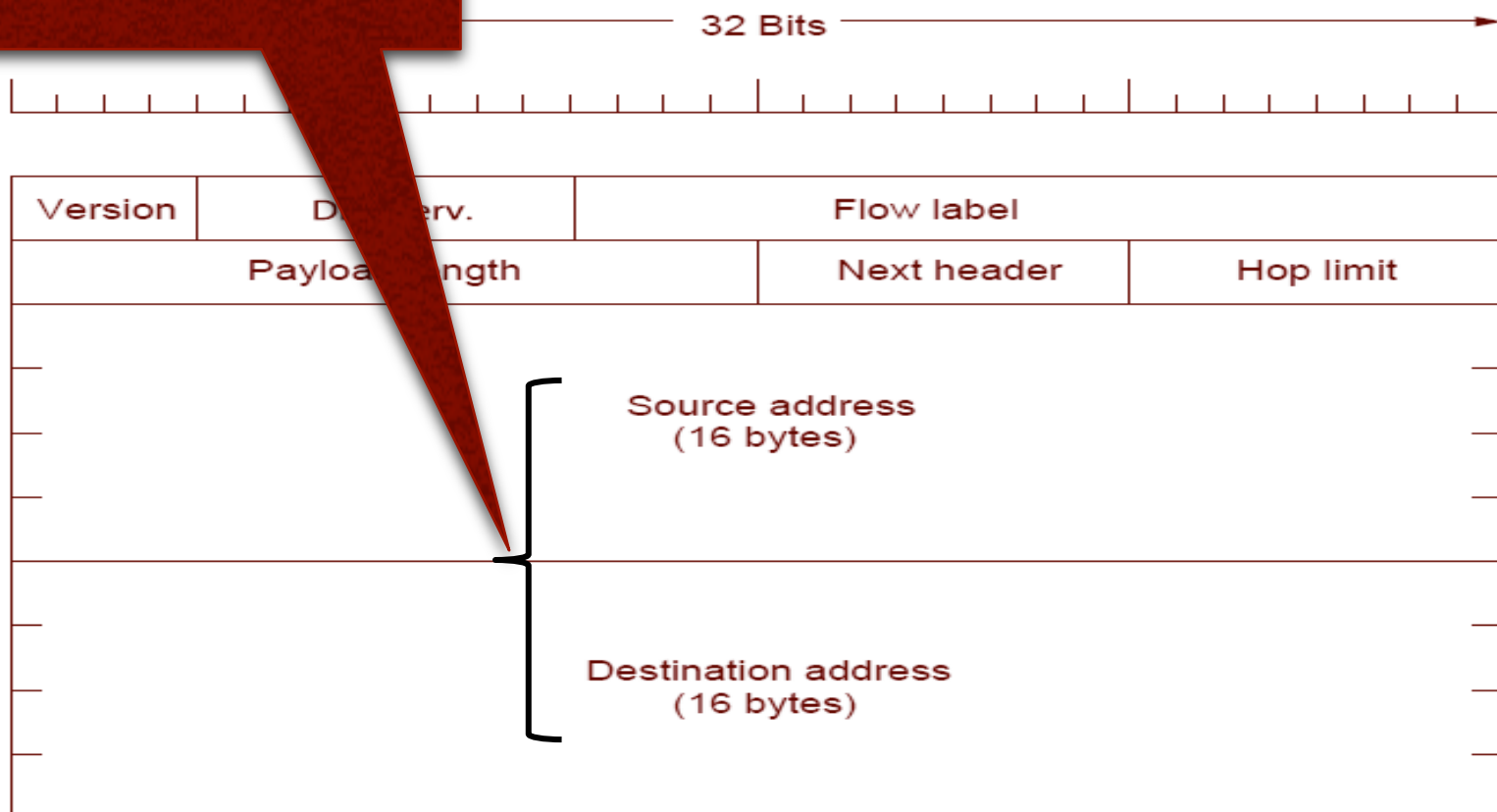
◆ *Required* header

*Limits packet lifetime
(same meaning as time to
live field in IPv4)*



IP version 6 (*IPv6*)

- ◆ Source & destination endpoint addresses



IP version 6 (*IPv6*)

- ◆ Extension headers handle other functionality

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents